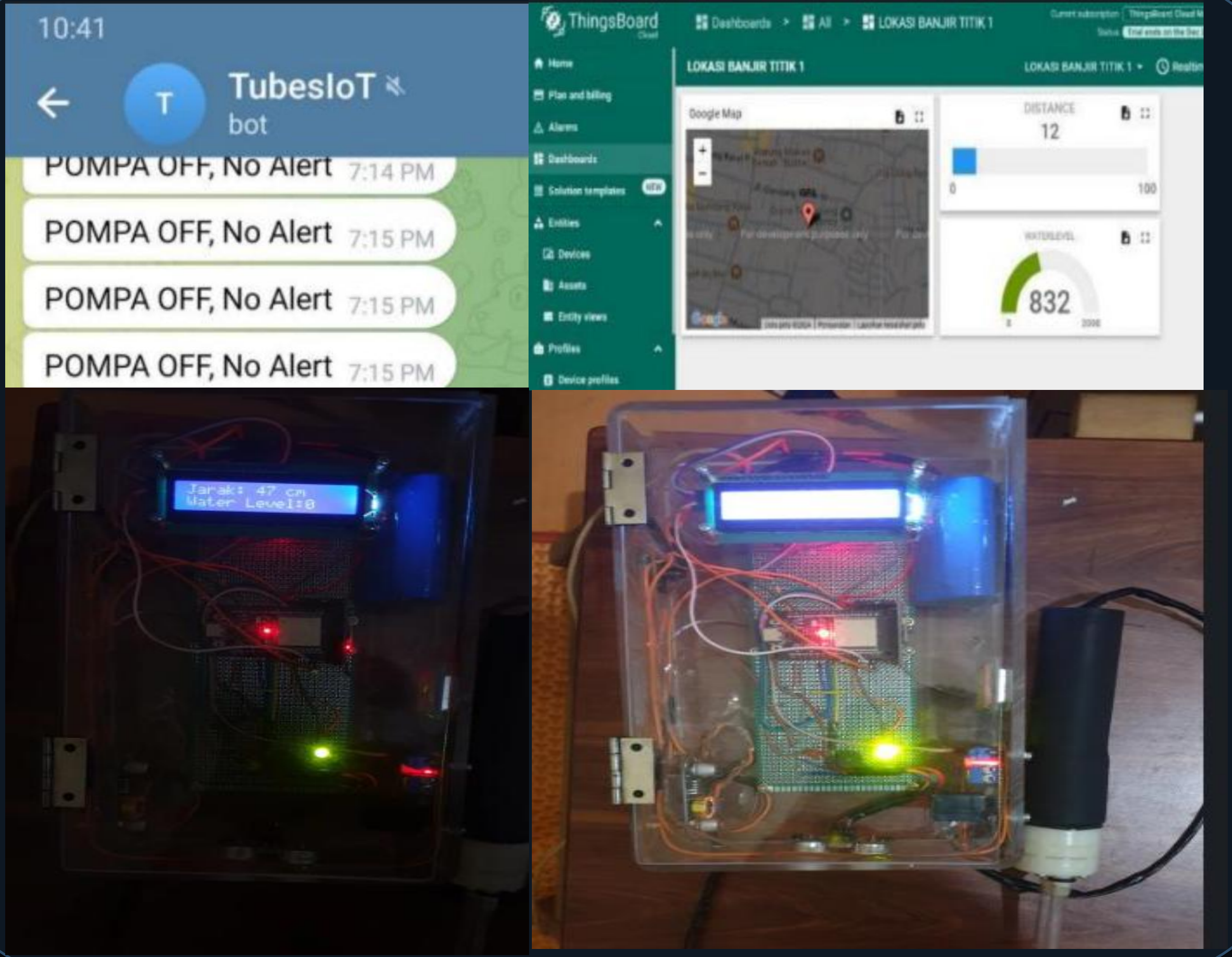
ESP32 FLOOD DETECTION & MITIGATION SYSTEM

NOV 2024 – DEC 2024 • UNIVERSITAS DIPONEGORO



PROJECT OVERVIEW

Real-time flood monitoring and mitigation

Designed an ESP32-based system to detect water-level conditions and support automated flood mitigation through sensors, GPS, actuator control, and Telegram notifications.

WATER LEVEL — Ultrasonic sensing for real-time water-level measurement.

GPS LOCATION — Provides location information for the monitored system.

ACTUATOR CONTROL — Supports automated mitigation through actuator/pump control.

TELEGRAM ALERT — Sends system status and flood-related notifications.

ESP32 • ULTRASONIC • GPS • ACTUATOR • TELEGRAM • THINGSBOARD

SYSTEM WORKFLOW

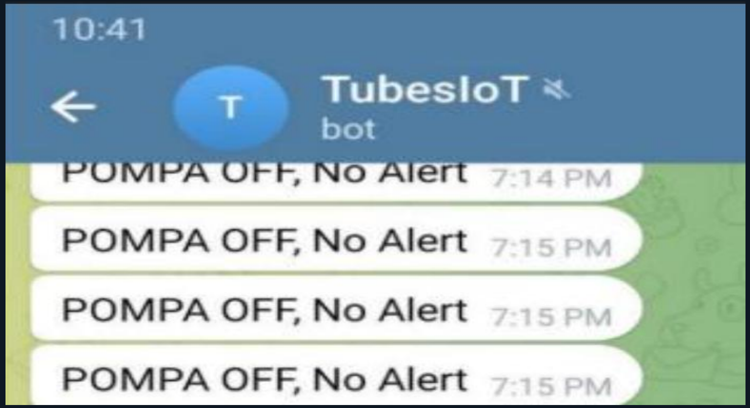
DETECTION → DECISION → MITIGATION → NOTIFICATION

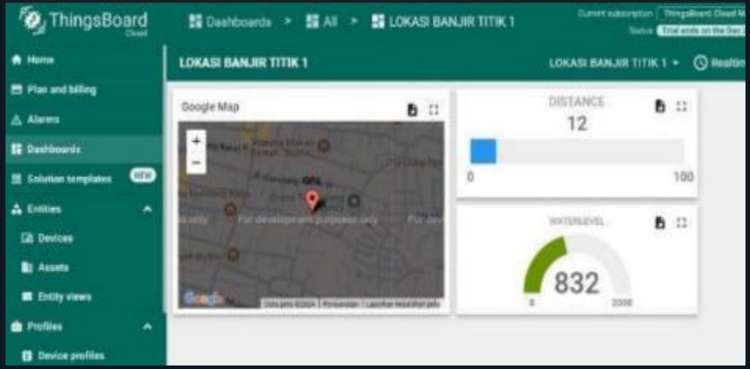
01 **DETECT**
Measure water level using the ultrasonic sensor.

02 **PROCESS**
ESP32 processes the measured condition and system state.

03 **MITIGATE**
Actuator/pump is controlled according to the system condition.

04 **NOTIFY**
Telegram provides status/alert information and the system location can be monitored.





KEY TECHNOLOGIES

ESP32 • Ultrasonic Sensor • GPS • Actuator • Telegram • ThingsBoard